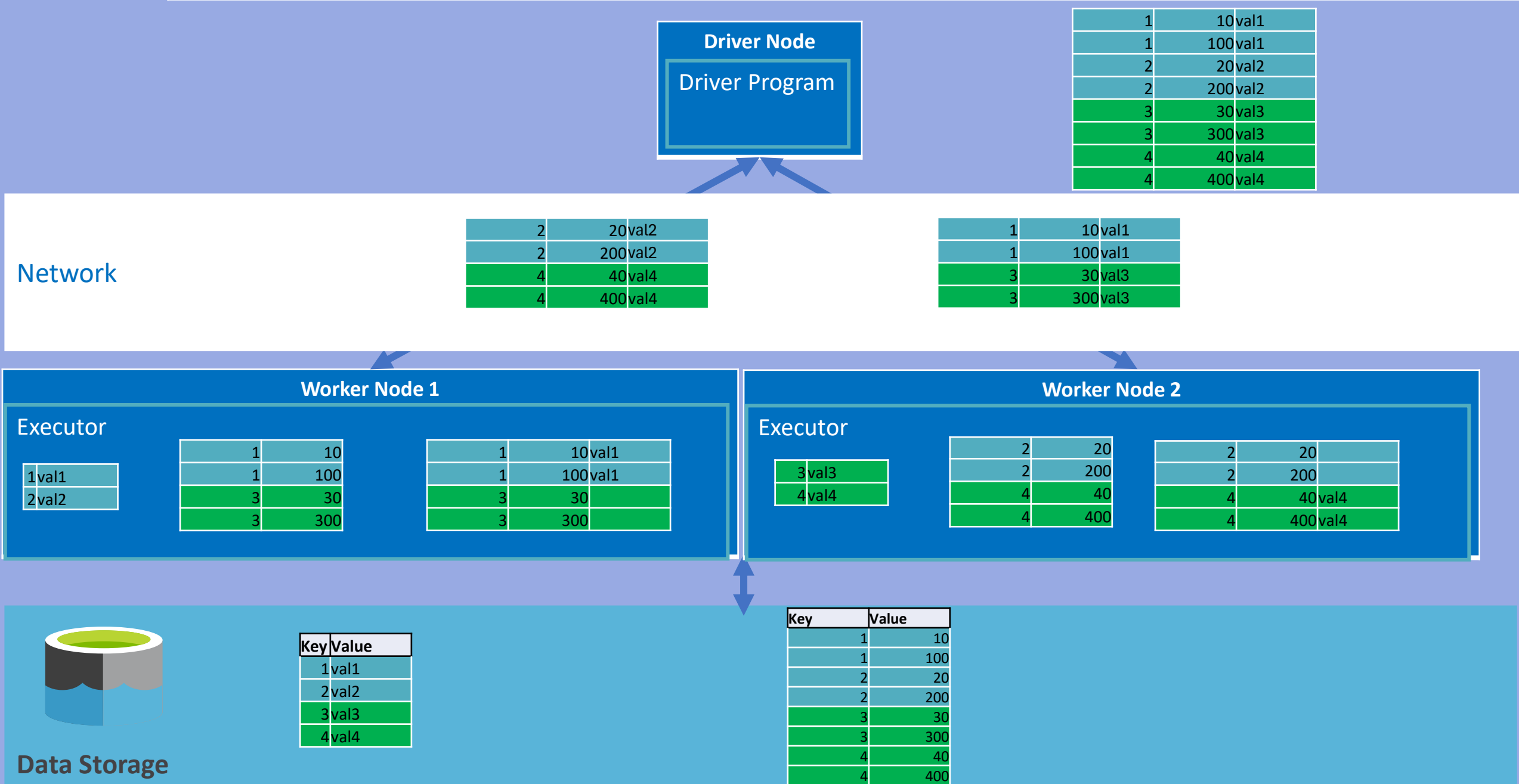
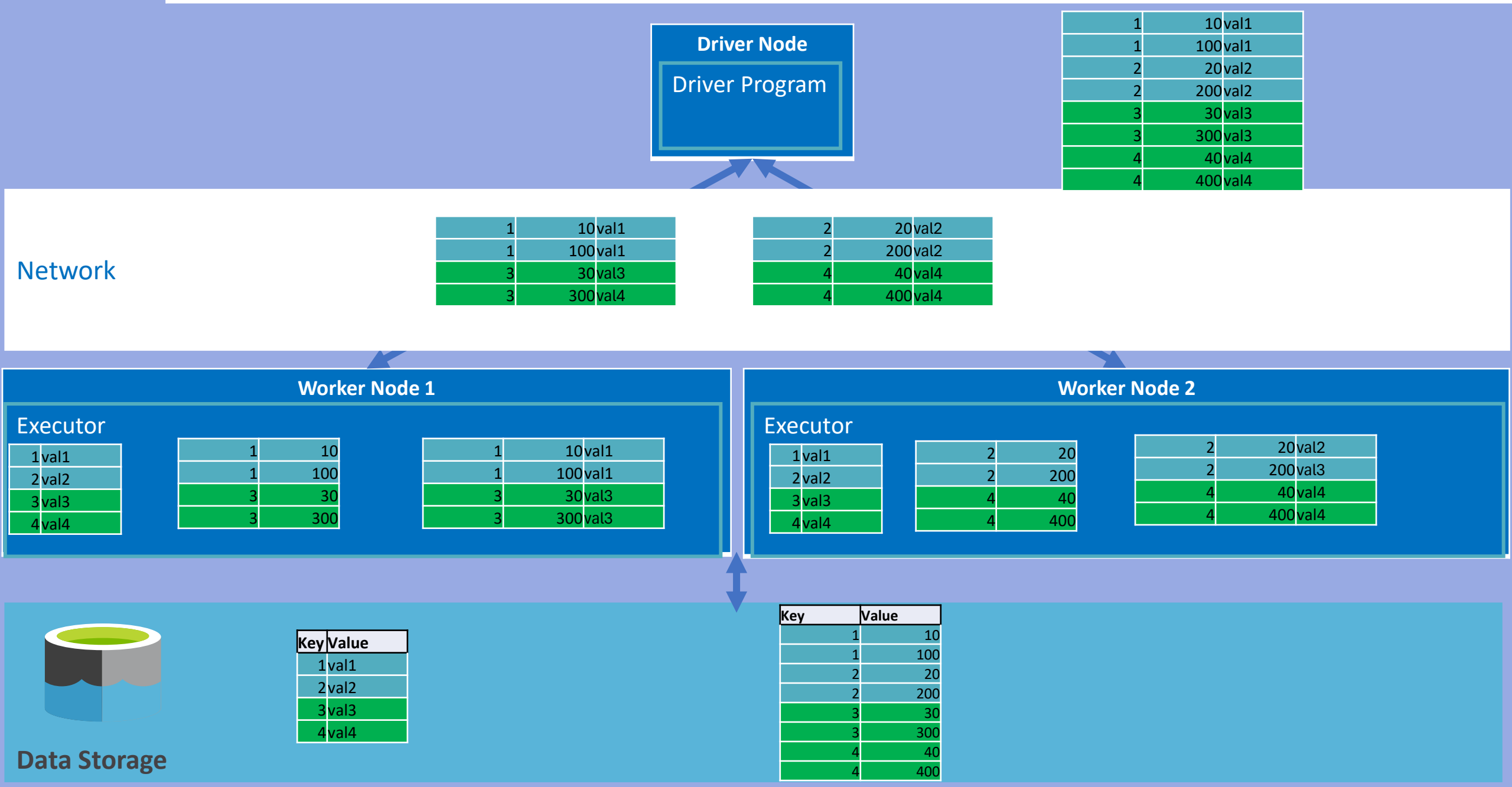


Performance Tuning – Data Shuffling Issue



Performance Tuning – Data Shuffling Broadcast Join



Performance Tuning – Data Shuffling SHUFFLE-HASH

Driver Node

Driver Program

Network

1	10	val1
1	100	val1
2	20	val2
2	200	val2
3	30	val3
3	300	val3
4	40	val4
4	400	val4

Worker Node 1

Executor

2	20
2	200

1	10
1	100
2	20
2	200

1	10	val1
1	100	val1
2	20	val2
2	200	val2

Worker Node 2

Executor

3	val3
4	val4

3	30
3	300
4	40
4	400

3	30	val3
3	300	val3
4	40	val4
4	400	val4

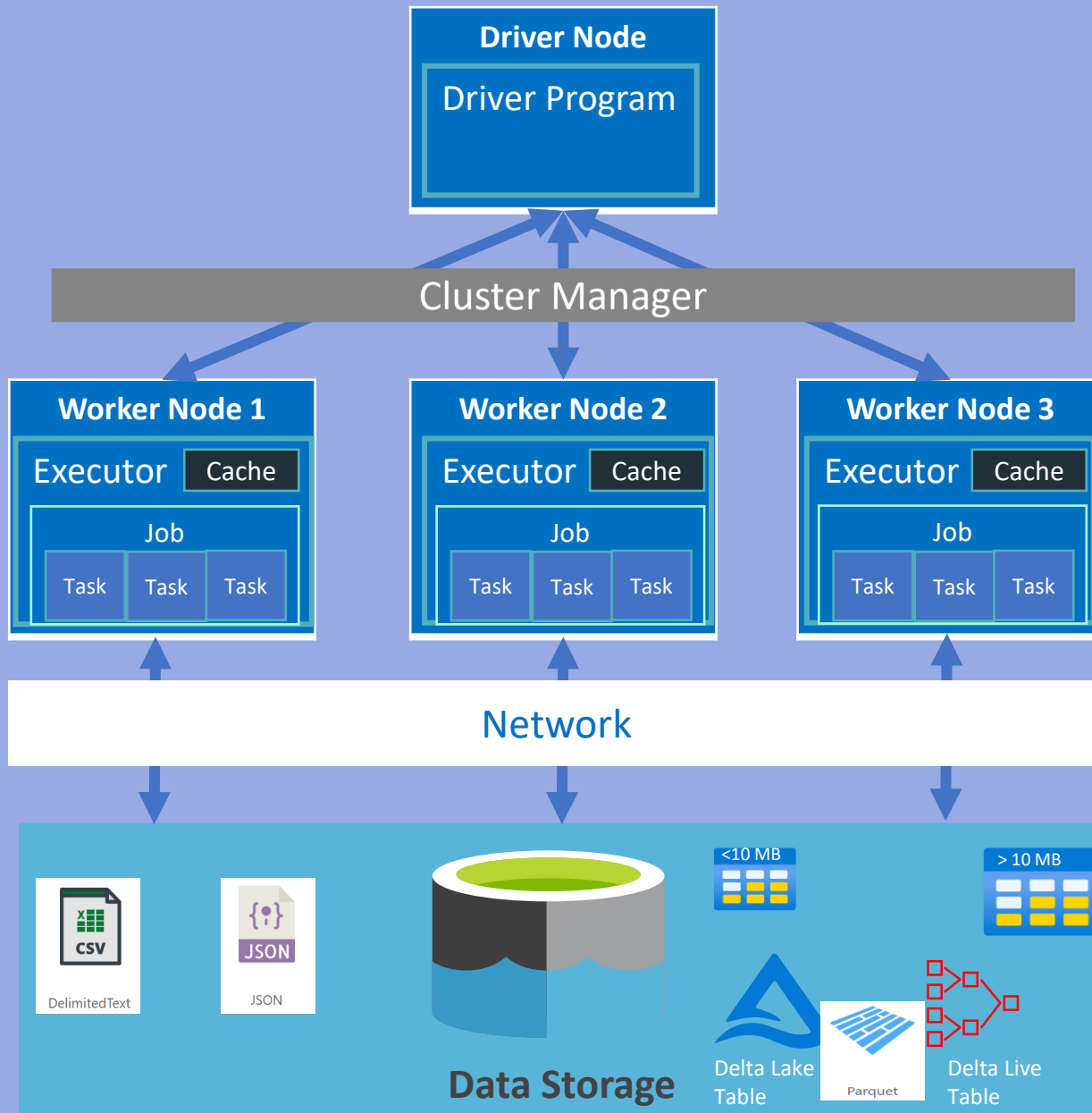


Data Storage

Key	Value
1	val1
2	val2
3	val3
4	val4

Key	Value
1	10
1	100
2	20
2	200
3	30
3	300
4	40
4	400

Performance Tuning – Data Shuffling Issue



Use **BROADCAST HASH JOIN(BHJ)** on small tables to copy all table data in all workers (Driver Node Does this task)
Spark Automatically **BROADCAST TABLES** size less than **10MB** , Adjust this size set `spark.sql.autoBroadcastJoinThreshold = <size in bytes>`

Use SQL HINTS to specify **BROADCAST** , **LEFT OUTER JOIN** Only Supported
`SELECT /*+ BROADCAST(dateDim) */ * FROM pricing_analytics.gold.reporting_dim_date_gold dateDim`

AQE(Adaptive Query Execution) converts all **SORT_MERGE** join into **BROADCAST HASH JOIN(BHJ)** for tables size less than 30 MB, Set this size using `set spark.databricks.adaptive.autoBroadcastJoinThreshold = <size in bytes>`

Driver node copies the data into Worker nodes and if we got larger Driver Node Memory (32 GB+) then we can set Broadcast table size up to 200 MB

Use **SHUFFLE-HASH** Join instead of **SORT-MERGE** join by setting the property `set spark.sql.join.preferSortMergeJoin = false` , **Photon Engine** replaces **SORT_MERGE** join with **SHUFFLE-HASH** Join

Use **COST-BASED-OPTIMIZER(CBO)** for queries with multiple Joins, use property `set spark.sql.cbo.enabled = false` (default true).
For effective CBO usage tables need to be **ANALYZED** regularly to collect statistics

`ANALYZE TABLE pricing_analytics.gold.reporting_fact_daily_pricing_gold COMPUTE STATISTICS FOR COLUMNS STATE_ID, MARKET_ID, PRODUCT_ID`

Re-order Join execution when using multiple joins in one SQL Query and use `set spark.sql.cbo.joinReorder.enabled = true`
ANALYZE table regularly is important for CBO & JOIN RE-ORDER work effective